

Coding & Solution

Date	15 March 2026
Team ID	XXXXXX
Project Name	Comprehensive measure of Well-Being
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Ishwaryaketha/HDI-Predictor
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	HDI Prediction using Linear Regression • Data Preprocessing and Cleaning • Flask Web Application • Analysis Dashboard • Heatmap, Scatter Plot, Distribution Plot, Pair Plot, Strip Plot • About Page • Model Serialization using Pickle
Pending / Incomplete Features	• User Authentication (Future Enhancement) • Cloud Deployment (Future Enhancement) • Country Comparison Dashboard
Setup / Run Instructions	1. Clone the GitHub repository. 2. Install dependencies using <code>pip install -r requirements.txt</code> . 3. Run <code>python model.py</code> to train the model (if needed). 4. Run <code>python app.py</code> . 5. Open <code>http://127.0.0.1:5000</code> in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

The HDI Predictor is developed using Python and Flask following a modular architecture. The application performs data preprocessing, trains a Linear Regression model, predicts the Human Development Index (HDI), and visualizes the results using Matplotlib and Seaborn. The project uses a CSV dataset and a Pickle file (`model.pkl`) for efficient prediction. The code is organized into reusable modules, making it easy to maintain and extend with additional machine learning models or features in the future.